

- 7** A student mixes 25.0 cm^3 of $0.350 \text{ mol dm}^{-3}$ sodium hydroxide solution with 25.0 cm^3 of $0.350 \text{ mol dm}^{-3}$ hydrochloric acid. The temperature increases by $2.5 \text{ }^\circ\text{C}$. No heat is lost to the surroundings.

The final mixture has a specific heat capacity of $4.2 \text{ J g}^{-1} \text{ K}^{-1}$.

What is the molar enthalpy change for the reaction?

- A** -150 kJ mol^{-1}
- B** -60 kJ mol^{-1}
- C** -30 kJ mol^{-1}
- D** $-0.15 \text{ kJ mol}^{-1}$